FIITJEE

ALL INDIA TEST SERIES

FULL TEST – I

JEE (Main)-2020

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 75 questions.
- Each subject (PCM) has 25 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each part has only three sections: **Section-A**, **Section-B and Section-C**.

  Section-A (01 – 20, 26 – 45, 51 – 70) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 22, 46 – 47, 71 – 72) *contains 6 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

  Section-C (23 – 25, 48 – 50, 73 – 75) *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

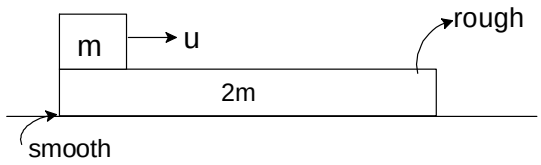
Physics

PART – I

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. The total work done by friction (assuming plank is sufficiently long) is:



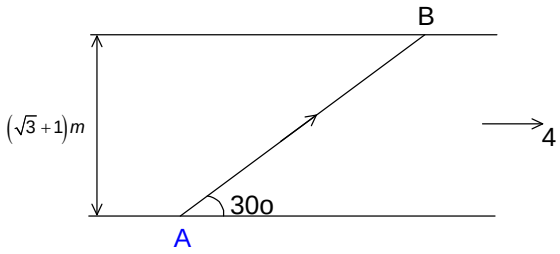
(A) $\frac{1}{3}mu^2$

(B) $-\frac{1}{3}mu^2$

(C) $\frac{3mu^2}{18}$

(D) $-\frac{3mu^2}{18}$

2. A person has to cross the river along the path shown, velocity of river flow is 4 m/s. Speed of person in still water is $2\sqrt{2}$ m/s. Time taken by the person to cross the river may be



(A) 2 s

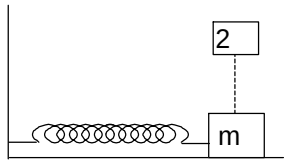
(B) $\left(2-\sqrt{3}\right)s$

(C) $\left(2+\sqrt{3}\right)s$

(D) $\left(1+\frac{\sqrt{3}}{2}\right)s$

Space for Rough work

3. An object of mass 'm' is performing SHM on a smooth horizontal surface. When the object reaches extreme position, another object of mass '2m' is gently placed on the oscillating object and sticks to it. The correct statements are



(A) amplitude of oscillation will change
(B) time period of oscillation remains unchanged
(C) total mechanical energy of the system remains constant
(D) maximum speed of the oscillating object does not change

4. Maximum spectral radiancy of a black body corresponds to 'λ' wavelength. If temperature is now changed so that maximum spectral radiancy now corresponds to $\frac{3\lambda}{4}$ then;

(A) new temperature is $\frac{3}{4}$ times of the old temperature

(B) new temperature is $\frac{4}{3}$ times of the old temperature

(C) Power radiated by the body changes by a factor of $\frac{100}{81}$

(D) power radiated by the body changes by a factor of $\frac{81}{256}$

5. Two equal spheres A & B lie on a smooth horizontal circular groove at opposite ends of a diameter. Sphere A is projected along the groove and at the end of time 'T' impinges on sphere B. If 'e' is the coefficient of restitution, the second impact will occur after a time equal to

(A) T (B) eT

(C) $\frac{2T}{e}$ (D) 2eT

6. Two bodies of masses M_1 = m and M_2 = 4 m are placed at a distance r. The gravitational potential at a point on the line joining them where the gravitational field is zero is

(A) zero (B) $-\frac{4Gm}{r}$

(C) $-\frac{6Gm}{r}$ (D) $-\frac{9Gm}{r}$

Space for Rough work

7. A solid metallic sphere of radius R having moment of inertia equal to I about its diameter is melted and recast into a solid disc of radius r of uniform thickness. The moment of inertia of the disc about an axis passing through its edge and perpendicular to its plane is also equal to I. The ratio $\left(\frac{r}{R}\right)$ is

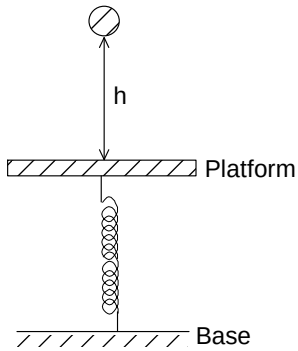
(A) $\frac{2}{\sqrt{15}}$ (B) $\frac{2}{\sqrt{10}}$

(C) $\frac{2}{\sqrt{5}}$ (D) $\frac{1}{\sqrt{2}}$

8. A body of mass m = 1 kg is dropped from a height h = 40 cm on a horizontal platform fixed to one end of an elastic spring, the other being fixed to a base, as shown. As a result the spring is compressed by an amount x = 10 cm. What is the force constant of the spring. ($g = 10\ ms^{-2}$).

(A) 600 N m^{-1} (B) 800 N m^{-1}

(C) 1000 N m^{-1} (D) 1200 N m^{-1}

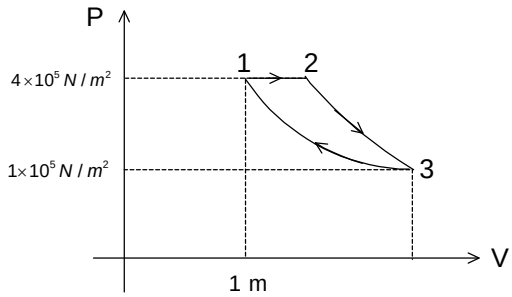


9. An organ pipe, open at the both ends and another organ pipe, closed at one end, will resonate with each other, if their lengths are in the ratio of

(A) 1 : 1 (B) 1 : 4

(C) 2 : 1 (D) 1 : 2

10. A system undergoes three quasi-static processes sequentially. 1 – 2 is an isobaric process, 2 – 3 is a polytropic process with $\gamma = \frac{4}{3}$ and 3 – 1 is an isothermal process. The heat transfer for the cycle is ΔQ, the change in internal energy is ΔU an the work done is ΔW. Then



(A) $\Delta W = 0$

(B) $\Delta Q = 10^5$ J

(C) $\Delta U = 0$

(D) $\Delta Q > \Delta W$

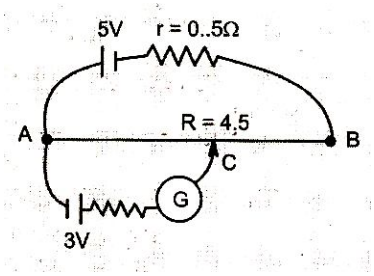
Space for Rough work

11. The density of a solid at normal pressure is ρ when the solid is subjected to an excess pressure P, the density changes to ρ'. It the bulk modulus of the solid is K then the ratio $\frac{\rho'}{\rho}$ is approximately

(A) $1+\frac{\rho}{K}$ (B) $1-\frac{\rho}{K}$

(C) $2+\frac{\rho}{K}$ (D) $z-\frac{\rho}{K}$

12. In the given potentiometer circuit length of the wire AB is 3 m and resistance is R = 4.5Ω. The length AC for no deflection in galvanometer is :
(A) 2 m
(B) 1.8 m
(C) dependent on r_1
(D) None of these

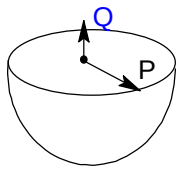


13. A point charge $q = 50\mu C$ is located in the x – y plane at the point of position vector $\vec{r_o} = 2\hat{i} + 3j$. What is the electric field at the point of position vector $\vec{r} = 8\hat{i} - 5j$

(A) $1200\frac{v}{m}$ (B) $4\times10^{-2}\frac{v}{m}$

(C) $900\frac{v}{m}$ (D) $4500\frac{v}{m}$

14. A point charge Q is located just above the centre of the flat face of hemisphere as shown in figure. The electric flux through the flat face and curved face of hemisphere are respectively

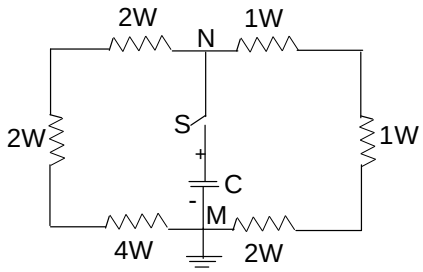


(A) $\frac{Q}{2\epsilon_o}, \frac{-Q}{2\epsilon_o}$ (B) $\frac{-Q}{2\epsilon_o}, \frac{Q}{2\epsilon_o}$

(C) $\frac{Q}{\epsilon_o}, \frac{-Q}{\epsilon_o}$ (D) $\frac{-Q}{\epsilon_o}, \frac{Q}{\epsilon_o}$

Space for Rough work

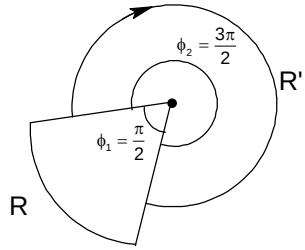
15. A capacitor of capacity 6μF and initial charge 160μC is vconnected with key S and resistance as shown in figure point M is earthed if key is closed at t = 0 then the current through resistance R $(=1\Omega)$ at t = 16μs is



(A) $\frac{10}{3e}$A (B) $\frac{10}{e}$A

(C) $\frac{20}{3e}$A (D) None

16. A current of I ampere is flowing through each of the bent wires as shown. The magnitude and direction of magnetic field at O is.



(A) $\frac{\mu_o i}{4}\left(\frac{1}{R}+\frac{2}{R}\right)$ (B) $\frac{\mu_o i}{4}\left(\frac{1}{R}+\frac{3}{R}\right)$

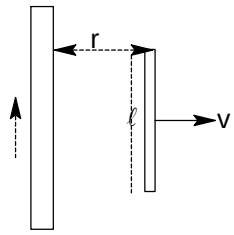
(C) $\frac{\mu_o i}{8}\left(\frac{1}{R}+\frac{3}{2R}\right)$ (D) $\frac{\mu_o i}{8}\left(\frac{1}{R}+\frac{3}{R}\right)$

17. In the arrangement shown in given figure current from A to B is increasing in magnitude induced current in the loop will



(A) have clockwise direction
(B) have anticlockwise direction
(C) be zero
(D) oscillate between clockwise and anticlockwise

18. A conducting rod moves with constant velocity v perpendicular to the long, straight wire carrying a current I as shown compute that the emf generated between the ends of the rod.



(A) $\frac{\mu_o v I \ell}{\pi r}$ (B) $\frac{\mu_o v I \ell}{2\pi r}$

(C) $\frac{2\mu_o v I \ell}{\pi r}$ (D) $\frac{\mu_o v I \ell}{4\pi r}$

Space for Rough work

19. A charged particle goes undeflected in a region containing electric and magnetite fields. It is possible that:

(A) $\vec{E} \parallel \vec{B}, \vec{V} \parallel \vec{E}$ (B) $\vec{E}$ is parallel to $\vec{B}$

(C) $\vec{V} \parallel \vec{B}$ but $\vec{E}$ is not parallel to $\vec{B}$ (D) $\vec{E} \parallel \vec{B}$ but $\vec{V}$ is not parallel to $\vec{E}$

20. When the electron in a hydrogen atom makes the transition $n_1 \rightarrow n_2$, the time period of the electron in its orbit becomes one – eighth of the original. Then (n_1, n_2) can be

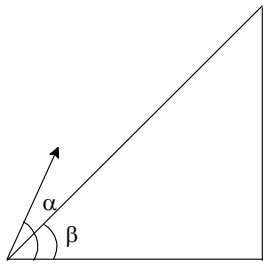
(A) (4, 2) (B) (8, 2)

(C) (8, 1) (D) (4, 3)

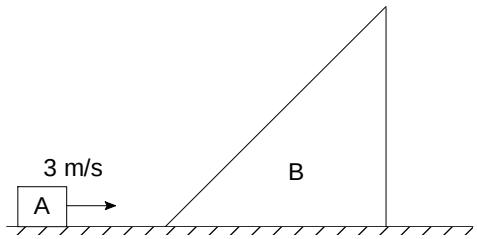
SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

21. A particle is projected up an inclined plane of inclination β to be horizontal as shown. The angle of projection of particle with horizontal is α. If the particle strikes the plane horizontally, then $\tan\beta = \frac{\tan\alpha}{x}$, find the value of x.



22. In the figure shown A is of mass 1 kg and B of mass 2 kg. A moves with velocity 3 m/s and rises on B which is initially at rest. All the surfaces are smooth. By the time A reaches the highest point on B, find the work done by A on B in joule.



Space for Rough work

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

23. Three wave forms travelling along a straight line have the forms $2A\sin\left(Kx-\omega t+\frac{\pi}{3}\right)$, $\sqrt{3}A\cos\left(Kx-\omega t-\frac{\pi}{3}\right)$, $2\sqrt{3}A\cos\left(Kx-\omega t+\frac{\pi}{3}\right)$. If the amplitude of the resulting wave front is $\sqrt{\frac{95}{x}}A$ unit then X is

24. Two thin symmetrical lenses of different nature have equal radii of curvature of all faces R = 20 cm. The lenses are put close together and immersed in water. The focal length of the system is 24 cm. The difference between refractive indices of the lenses is $\left(\frac{5}{n}\right)$. Find n. The refractive index of water is $\frac{4}{3}$.

25. Sounds from 2 identical sources S_1 & S_2 reach a point P. When the sounds reach in phase, the intensity at P is I_o. The power of S_1 is now reduced by 64% and the phase difference between S_1 & S_2 is varied continuously. The maximum and minimum intensities recorded at P are now I_{max} and I_{min}, then the value of $\frac{I_{max}}{I_{min}}$

Space for Rough work

Chemistry

PART – II

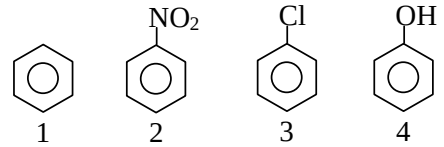
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

26. Diamond exists in FCC unit cell with 50% tetrahedral sites also occupied. The effective number of carbon atoms in a unit cell of diamond is
(A) 4 (B) 6
(C) 8 (D) 12

27. Van Arkel method of purification of metals involves converting the metal to a:
(A) volatile stable compound. (B) non-volatile stable compound.
(C) volatile unstable compound. (D) non-volatile unstable compound.

28. Identify the correct order of reactivity in electrophilic substitution reaction of the following compounds



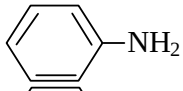
(A) 1 > 2 > 3 > 4 (B) 4 > 1 > 2 > 3
(C) 4 > 3 > 2 > 1 (D) 4 > 1 > 3 > 2

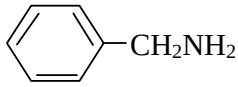
29. The reaction, $\frac{1}{2}H_2(g) + AgCl(s) \longrightarrow H^+(aq) + Cl^-(aq) + Ag(s)$, occurs in the galvanic cell:
(A) Ag(s) | AgCl(s) | KCl (solution) || $AgNO_3$(solution) | Ag(s)
(B) Pt | H_2(g) | HCl (solution) || $AgNO_3$(solution) | Ag(s)
(C) Pt | H_2(g) | HCl (solution) || AgCl(s) | Ag(s)
(D) Pt | H_2(g) | KCl (solution) || AgCl(s) | Ag(s)

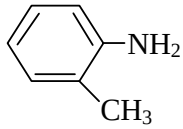
30. Which of the following will have highest coagulating power for As_2S_3 colloid?
(A) PO_4^{3-} (B) SO_4^{2-}
(C) Al^{3+} (D) Na^+

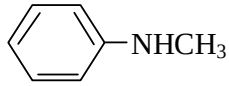
Space for Rough work

31. Which one of the following amines is the weakest base?

(A) 

(B) 

(C) 

(D) 

32. Which of the following is not soluble in NaOH?
(A) $Fe(OH)_3$ (B) $Zn(OH)_2$
(C) $Al(OH)_3$ (D) $Sn(OH)_2$

33. Which one of the following polymers may be classified as step growth polymer?
(A) Teflon (B) Polythene
(C) PVC (D) Nylon 66

34. At 550 K, the K_c for the following reaction is 10^4 mol^{-1} L

$X(g) + Y(g) \rightleftharpoons Z(g)$

At equilibrium, it was observed that

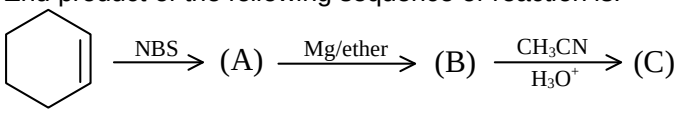
$$[X] = \frac{1}{2}[Y] = \frac{1}{2}[Z]$$

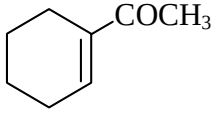
What is the value of [Z] at equilibrium
(A) 2×10^{-4} mol L^{-1} (B) 10^{-4} mol L^{-1}
(C) 2×10^{4} mol L^{-1} (D) 10^{4} mol L^{-1}

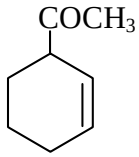
35. Which one of the following solutions in water will have the highest freezing point?
(A) Solution which is 0.1 M in KNO_3 and 0.2 M in $Ba(NO_3)_2$.
(B) Solution which is 0.1 M in $Ca(NO_3)_2$ and 0.1 M $Ba(NO_3)_2$.
(C) Solution which is 0.2 M in urea and 0.2 M glucose.
(D) Solution which is 0.1 M both in $Al_2(SO_4)_3$ and 0.1 M K_2SO_4.

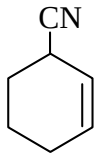
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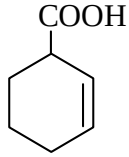
36. End product of the following sequence of reaction is:



(A) 

(B) 

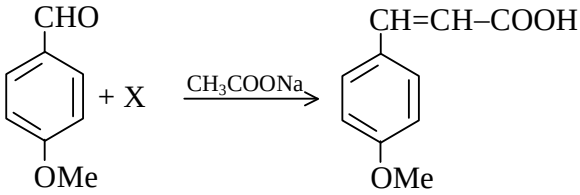
(C) 

(D) 

37. Which one of the following species is paramagnetic?

(A) O_2 (B) CN^-

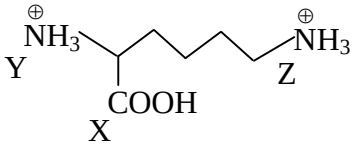
(C) CO (D) NO^+

38. 

what is X?

(A) CH_3COOH (B) $BrCH_2COOH$

(C) $(CH_3CO)_2O$ (D) CHO—COOH

39. 

the acid strengths are:

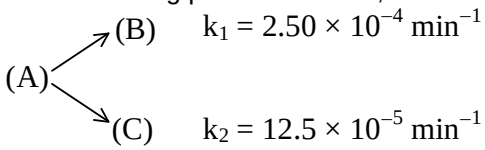
(A) X > Y > Z (B) X > Z > Y

(C) Y > Z > X (D) Z > X > Y

Space for Rough work

40. The number of P–O–P bonds in cyclic trimetaphosphoric acid is
(A) Zero (B) Two
(C) Three (D) Four

41. In the following parallel reaction,

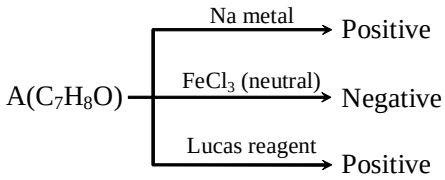


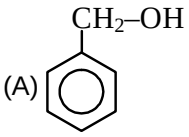
The percentage of product C formed at any time is
(A) 66.6% (B) 50%
(C) 33.3% (D) 12.5%

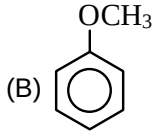
42. Which of the following ions is responsible for the brown colour in the ring test for a nitrate?
(A) $[Fe(H_2O)_5NO]^{2+}$ (B) $[Fe(CN)_5NO]^{2-}$
(C) $[Fe(NO_2)_6]^{4-}$ (D) $[Fe(H_2O)_5NO_2]^{+}$

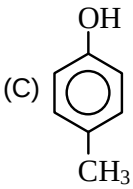
43. Which pair of the gaseous species diffuses through a small jet with the same rate of diffusion at same P and T?
(A) NO and CO (B) NO and CO_2
(C) NH_3 and PH_3 (D) NO and C_2H_6

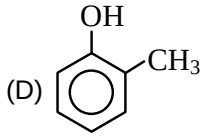
44. Identify an aromatic compound A(C_7H_8O) which gives following tests with the given reagents.











Space for Rough work

45. Identify the unknown products (A) and (B) in the given reaction:

Glucose + 3 $C_6H_5NH–NH_2 \xrightarrow{H^{\oplus}/\Delta}$ Osazone + (A) + (B)

(A) $C_6H_5NH_2$ and NH_3 (B) $C_6H_5NH_2$ and NH_2OH

(C) $C_6H_5NH—NHOH$ and NH_3 (D) NH_2OH and H_2O

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

46. In phosphorus atom (Z = 15) in ground state, the difference in number of occupied orbitals and occupied sub-shells are _____

47. The number of geometrical isomers for $[Pt(NH_3)_2Cl_2]$ is ____

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

48. 0.535 g mixture of ethanol and acetaldehyde when heated with Fehling solution gave 1.2 g of a red precipitate. What is the percentage of acetaldehyde in the mixture?

49. The relative rate of abstraction per hydrogen atom of 3°H, 2°H and 1°H of alkane during photo chlorination is 5.0 : 3.8 : 1. Determine the percentage of the major product formed in the chlorination of 2,2–dimethyl butane.

50. The pK_b value of ammonium hydroxide is 4.75. An aqueous solution of ammonium hydroxide is titrated with HCl. The pH of the solution at the point where half of ammonium hydroxide has been neutralised will be _____

Space for Rough work

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

51. If $|z| = Max.\{|z-1|, |z+1|\}$, then

(A) $|z+\bar{z}| = \frac{1}{2}$ (B) $z+\bar{z} = 1$

(C) $|z+\bar{z}| = 1$ (D) None of these

52. If x is real, then the maximum value of $3-6x-8x^2$ is

(A) $\frac{17}{8}$ (B) $\frac{33}{8}$

(C) $\frac{21}{8}$ (D) none

53. The maximum number of points into which 4 circles and 4 straight lines intersect is

(A) 26 (B) 50

(C) 56 (D) 72

54. $7^9 + 9^7$ is divisible by

(A) 16 (B) 24

(C) 64 (D) 72

55. The minimum no. of terms of the series 1 + 3 + 9 + 27 + …….. So that the sum may exceed 1000 is

(A) 7 (B) 5

(C) 3 (D) none of these

56. $\lim_{n\to\infty}\left(\cos\frac{x}{n}\right)^n$ is equal to

(A) e (B) e^{-1}

(C) 1 (D) none of these

Space for Rough work

57. Solution of the equation $\left(xy^4 + y\right)dx - xdy = 0$ is

(A) $4x^4y^3 + 3x^3 = cy^3$ (B) $3x^3y^4 + 4y^3 = cx^3$

(C) $3x^4y^3 + 4x^3 = cy^3$ (D) none

58. The point (1,β) lies on or inside the triangle formed by the lines y = x, x – axis and x + y = 8, if

(A) $0 < \beta < 1$ (B) $0 \le \beta \le 1$

(C) $0 < \beta < 8$ (D) none of these

59. From any point on the circle $x^2 + y^2 = a^2$ tangents are drawn to the circle $x^2 + y^2 = a^2 \sin^2 \alpha$. The angle between them is

(A) $\frac{\alpha}{2}$ (B) α

(C) 2α (D) none of these

60. The equation of common tangent to the curves $y^2 = 8x$ and xy = - 1 is

(A) 3y = 9x + 2 (B) y = 2x + 1

(C) 2y = x + 8 (D) y = x + 2

61. If $\vec{u} = \vec{a} - \vec{b}, \vec{v} = \vec{a} + \vec{b}$ and $\left|\vec{a}\right| = \left|\vec{b}\right| = 2$, then $\left|\vec{u} \times \vec{v}\right|$ =

(A) $2\sqrt{\left[16 - (a.b)^2\right]}$ (B) $\sqrt{\left[16 - (a.b)^2\right]}$

(C) $2\sqrt{\left[4 - (a.b)^2\right]}$ (D) $\sqrt{\left[4 - (a.b)^2\right]}$

62. If 25% of the items are less than 20 and 25% are more than 40, the quartile deviation is

(A) 20 (B) 30

(C) 40 (D) 10

63. The no. of solutions of the equation $\sin x = \cos 3x$ in [0,π] is

(A) 1 (B) 2

(C) 3 (D) 4

Space for Rough work

64. The function $f(x)=\sec\left[\log\left(x+\sqrt{1+x^2}\right)\right]$ is

(A) even (B) odd
(C) const. (D) none of these

65. If $\Delta=\begin{vmatrix}1 & 3\cos\theta & 1\\ \sin\theta & 1 & 3\cos\theta\\ 1 & \sin\theta & 1\end{vmatrix}$, then maximum value of Δ is

(A) 10 (B) 14
(C) 1 (D) none of these

66. If $g(x)=\left(x^2+2x+3\right)f(x), f(0)=5$ and $\lim_{x\to 0}\frac{f(x)-5}{x}=4$, then g'(0) is equal to

(A) 22 (B) 20
(C) 18 (D) none of these

67. $f(x)=1+[\cos x]x,\ in\ \ 0<x\le\frac{\pi}{2}$

(A) has a minimum value 0 (B) has a maximum value 2
(C) is continuous in $\left[0,\frac{\pi}{2}\right]$ (D) is not differentiable at $x=\frac{\pi}{2}$

68. $\int\frac{dx}{\cos x\sqrt{\cos 2x}}$ is equal to

(A) $\sin^{-1}(\tan x)+c$ (B) $\cos^{-1}(\tan x)+c$
(C) $\tan^{-1}(\sin x)+c$ (D) none of these

Space for Rough work

69. The value of the integral $\int_0^{2[x]} \left(x - [x]\right) dx$ is

(A) [x] (B) $\frac{1}{2}[x]$

(C) $3[x]$ (D) $2[x]$

70. Seven chits are numbered 1 to 7. Four are drawn one by one with replacement. The probability that the least number on any selected chit is 5, is

(A) $\left(\frac{2}{7}\right)^4$ (B) $4\left(\frac{2}{7}\right)^4$

(C) $\left(\frac{3}{7}\right)^3$ (D) none

SECTION – B
(Single digit integer type)

This section contains **02** questions. The answer to each question is a **single Digit integer** ranging from **0 to 9, both inclusive**.

71. The co-eff. of x in the expansion of $\left[\sqrt{1+x^2} - x\right]^{-1}$ in ascending power of x, when $|x| < 1$, is

72. If $x = 2 + 2^{\frac{2}{3}} + 2^{\frac{1}{3}}$, then the value of $x^3 - 6x^2 + 6x$ is

SECTION – C
(Numerical Answer Type)

This section contains **03** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

73. If f(x + y) = 2f(x).F(y) $\forall$ x, y where $f'(0)$ = 3 and f(4) = 2, then $f'(4)$ is equal to

74. A bag contains 10 mangoes out of which 4 are rotten, two mangoes are taken out together. If one of them is found to be good, then the probability that other is also good is

75. How many non-negative integral solutions does the equation 3x + y + z + w = 30 have?

Space for Rough work